STAT 4620 Final Project Report

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1. Introduction

Our goal is to model the relationship between workers' wages and a set of demographic and labor-market characteristics using the Wage dataset from the ISLR package. Understanding how factors such as age, education, job classification, year, and marital status contribute to wage outcomes is important for both labor economics research and practical workforce planning. The objective is to identify a model that is both interpretable and well-supported by the data, and to summarize insights about which variables most strongly influence wages and how these effects behave across demographic groups.

2. Data Description and Cleaning

2.1 Dataset Structure

The Wage dataset includes 3000 observations with 13 variables. Table 1 shows the description and measurement type for each variable.

Table 1: Variable Descriptions for Wage Dataset

Variable	Description	Type
X	Respondent ID (unique identifier)	Numerical: Discrete
year	Year that wage information was recorded	Numerical: Discrete
age	Age of worker	Numerical: Discrete
maritl	Marital status: 1. Never Married, 2. Married, 3. Widowed, 4. Divorced, 5. Separated	Categorical: Nominal
race	Race of worker: 1. White, 2. Black, 3. Asian, 4. Other	Categorical: Nominal
education	Highest education level: 1. < HS Grad, 2. HS Grad, 3. Some College, 4. College Grad, 5. Advanced Degree	Categorical: Ordinal
region	Region of country (mid-atlantic only)	Categorical: Nominal
jobclass	Job class: 1. Industrial, 2. Information	Categorical: Nominal
health	Self-reported health status: 1. <=Good, 2. >=Very Good	Categorical: Ordinal
health_ins	Health insurance status: 1. Yes, 2. No	Categorical: Binary
logwage	Log-transformed workers wage	Numerical: Continuous
wage	Workers raw wage	Numerical: Continuous
Resp	Mystery response variable	Numerical: Continuous

2.2 Data Cleaning & Preprocessing

All original variables are complete. The appended variable Resp contains missing values. To examine potential bias, we created an indicator variable marking whether Resp was missing. Comparing summary statistics of numerical predictors between missing and non-missing Resp cases showed similar mean values, suggesting that the missingness is likely missing at random (Table 2). Therefore, removing rows with missing Resp will not bias the analysis. We also checked for duplicate rows and found none. Thus, our final cleaned dataset excludes the identifier variable X and all rows with missing Resp.

Table 2: Mean of Numeric Variables by Missingness of Resp

missing	X	year	age	logwage	wage	Resp
FALSE	219006.4	2005.79	42.42	4.65	111.73	35.66
TRUE	212854.1	2005.62	42.15	4.67	110.65	NaN

2.3 Recoding Marital Status (Married vs. Unmarried)

Because the original maritl variable contains five categories (Never Married, Married, Widowed, Divorced, Separated), many of the categories have small counts and their economic interpretation is similar. Additionally, preliminary modeling and exploratory analysis showed that **the primary distinction appears between Married and all other groups**.

Therefore, for all EDA and modeling work moving forward, we create a binary variable:

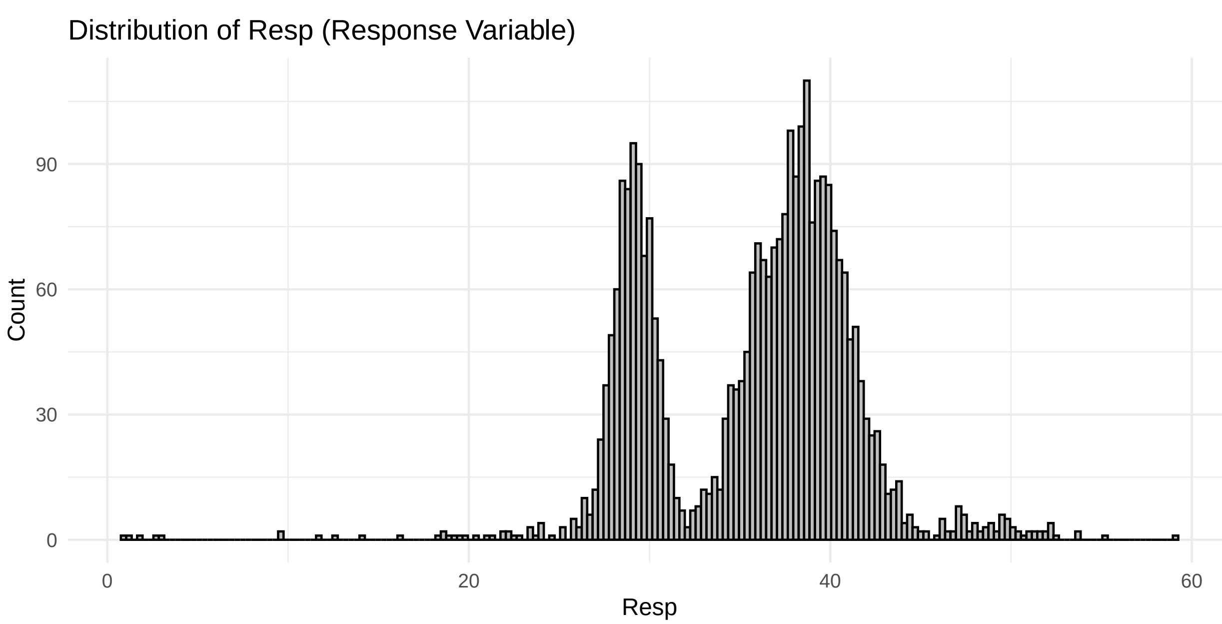
married2 = "Married" vs. "Unmarried."

This recoding makes interpretation easier and will later become important because marital status turns out to be the strongest interaction variable in both EDA and modeling.

3. Exploratory Data Analysis

3.1 Distribution of the Response Variable (Resp)

Before examining predictors, we examine the shape of the response variable. The histogram below shows that **Resp is clearly bimodal**, with two distinct peaks.



Interpretation

The strong **bimodal distribution** suggests that:

- A single linear mean structure may not fully capture the data.
- Some predictors may shift individuals into one mode or the other, particularly marital status.
- This helps explain why certain interactions appeared significant under linear modeling but not under more flexible models.

This motivates the later need for polynomial and semi-parametric models.

3.2 Summary of Numeric Variables & Categorical Variables

The variable wage shows a wide range, with a median that is lower than the mean, indicating the presence of some high earners. The variable logwage is the logarithmic transformation of wage, which reduces skewness and makes the distribution more symmetric (Table 3). The categorical variables were summarized by their most frequent categories (Table 4).

Table 3: Summary Statistics – Numeric Variables

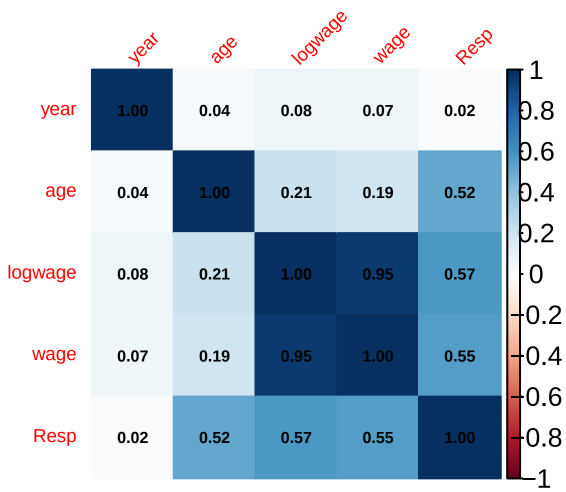
Variable	Min	Median	Mean	Max
year	2003.00	2006.00	2005.79	2009.00
age	18.00	42.00	42.42	80.00
logwage	3.00	4.65	4.65	5.76
wage	20.09	104.92	111.73	318.34
Resp	1.00	36.95	35.66	59.11

Table 4: Most Frequent Category – Categorical Variables

Variable	Most Frequent Category
maritl	2. Married
race	1. White
education	2. HS Grad
region	2. Middle Atlantic
jobclass	1. Industrial
health	2. >=Very Good
health_ins	1. Yes
married2	Married

3.3 Correlation Analysis

The variables wage and logwage are highly correlated, as expected due to the log transformation. The response variable Resp shows moderate positive correlations with wage, logwage, and age, suggesting that these variables may be influential in predicting Resp. The year variable shows a weak correlation with Resp, indicating limited time-based effects.

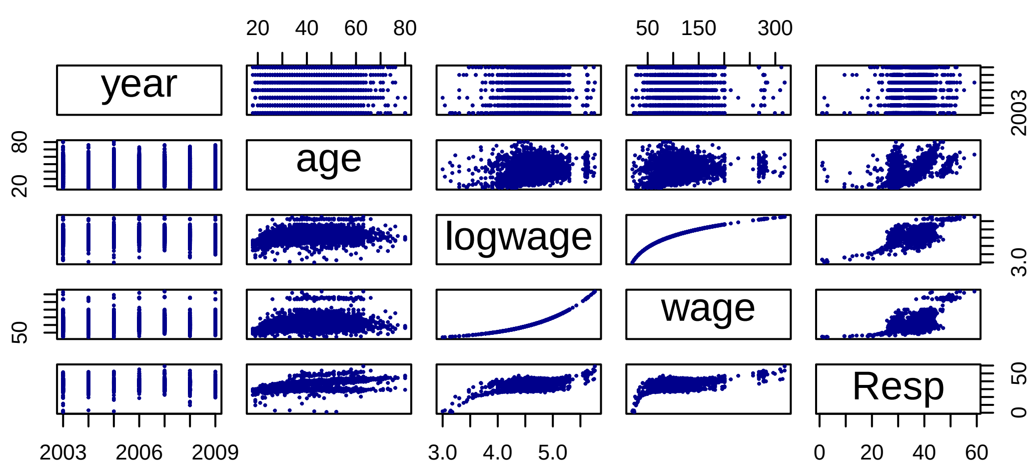


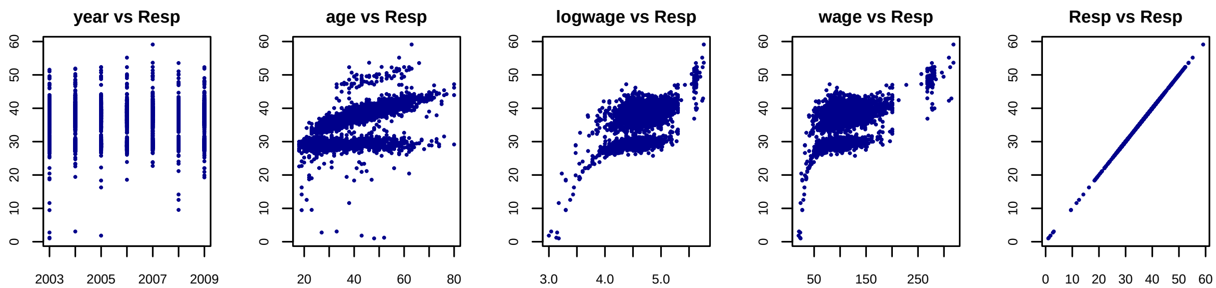
3.4 Numeric Variables vs Response

From scatterplot matrix, there is a positive, non-linear relationship. Wages tend to increase with age up to around 60, after which they seem to level off or slightly decrease. The variance also appears to increase with age. The plot also confirms that Resp is highly correlated with logwage and wage, and positively related to age.

When we specifically look at the relationship between Resp and the continuous predictors:

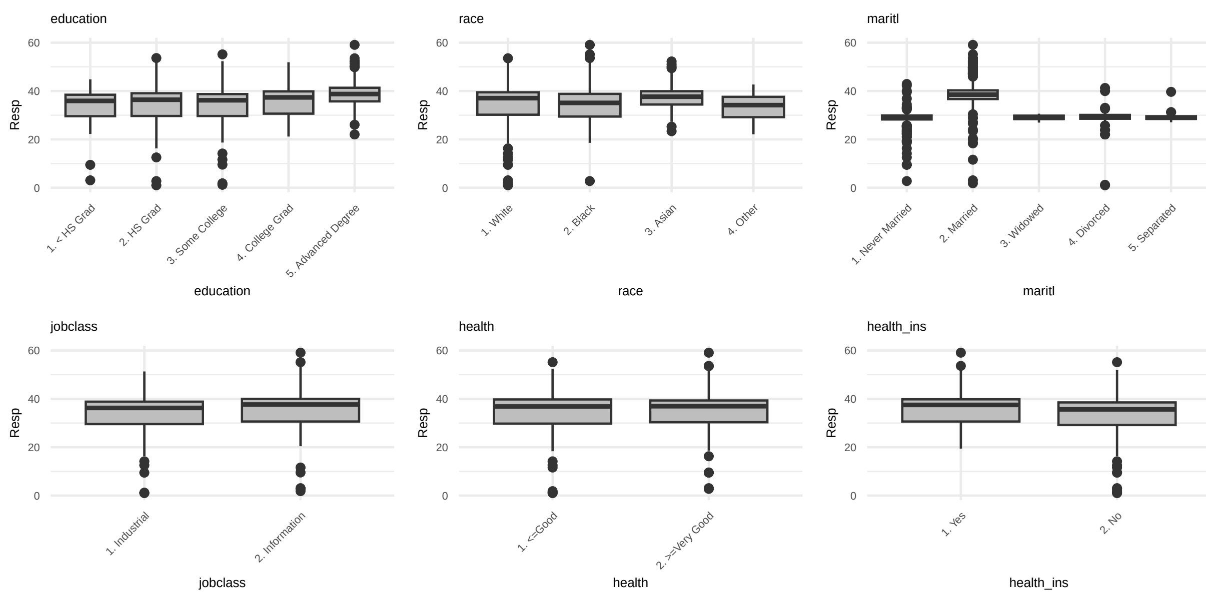
- age vs. Resp: The three apparent horizontal lines or clusters are most likely caused by the presence of a strong categorical predictor. A small number of highly influential categories in a predictor variable can create distinct, separated bands in a scatter plot when plotted against any continuous variable (like age).
- logwage/Wage vs. Resp: The plots are separated into distinct clusters primarily due to the influence of a strong categorical predictor.





3.5 Categorical Variables vs Response

- education: Clear, strong separation between levels. Higher education corresponds to higher Resp.
- maritl (Married vs Unmarried): This is one of the strongest relationships in the data. Married individuals disproportionately fall into the upper mode of the Resp distribution. This key observation explains the bimodal distribution and motivates the strong reliance on the married2 interaction term in the subsequent modeling sections.
- race, health, health_ins: These show weaker group separation. Their effects appear smaller and do not meaningfully explain the bimodality of Resp.
- jobclass: Practically no separation in Resp; likely not an important predictor.
- Outlier Identification: The outliers show data points with unusually low or high wages for their respective categories, which may warrant further investigation for potential influential observations.



3.6 Variables interations with Age and Logwage

Our earlier results suggested that both age and logwage are strong predictors of the response variable, so we examined whether their effects vary across different subgroups.

Interactions with Age

The interaction tests show clear evidence that the relationship between **age** and the response varies across several demographic and health-related variables. The strongest effect appears in the **age × maritl** interaction, followed by meaningful interactions with **health** and **health_ins**. These findings indicate that the effect of age is not uniform; instead, it depends on an individual's marital status, health condition, and insurance coverage—especially marital status. (Appendix 1)

Significant interactions with age:

- age × maritl (strongest)
- age × health
- age × health_ins

The interaction between age × maritl is highly significant ($p < 0.001$), confirming that marital status meaningfully alters the age effect on the response. Interactions with health and health_ins are also statistically significant, though to a lesser degree, indicating that age's effect varies in modest but important ways across these groups. (Appendix 3)

Table 5: ANOVA Interaction Summary

Interaction	F	p-value	Interpretation
age × maritl	69.738	< 2.2e-16	Highly significant.
age × health	8.324	0.00394	Significant.
age × health_ins	5.570	0.0183	Significant.

Interactions with Logwage

The interaction analysis for **logwage** shows a different pattern. The strongest and most meaningful interaction is with **health_ins**, indicating that the relationship between wages and the response variable differs considerably between individuals with and without health insurance. Marital status shows a marginally significant interaction, while education, jobclass, and health do not appear to moderate the logwage–response relationship. (Appendix 2)

Significant interactions with logwage:

- logwage × health_ins (strongest)

- logwage × maritl (marginal)

For logwage, the interaction with health_ins again emerges as highly significant ($p < 0.001$), showing that the wage effect differs sharply depending on health insurance status. The logwage × maritl interaction is marginally significant ($p = 0.047$), suggesting that while there may be small differences across marital groups, the effect is weaker. (Appendix 4)

Table 6: ANOVA Interaction Summary

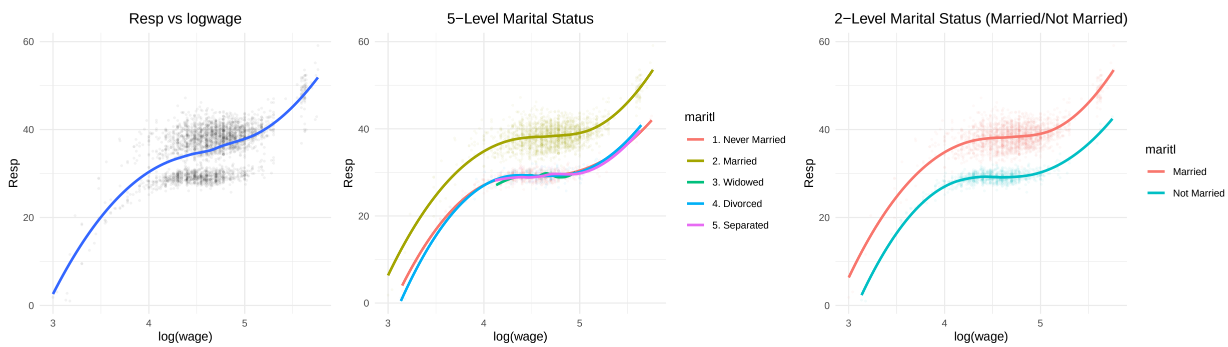
Interaction	F	p-value	Interpretation
logwage × health_ins	18.858	1.456e-05	Interaction significantly improves the model.
logwage × maritl	2.416	0.04675	Interaction barely significant.

Our exploratory interaction tests confirmed that the effect of continuous predictors is not uniform. The most critical interaction, which became the cornerstone of our best models, is the one involving marital status.

3.8 Understanding Interactions

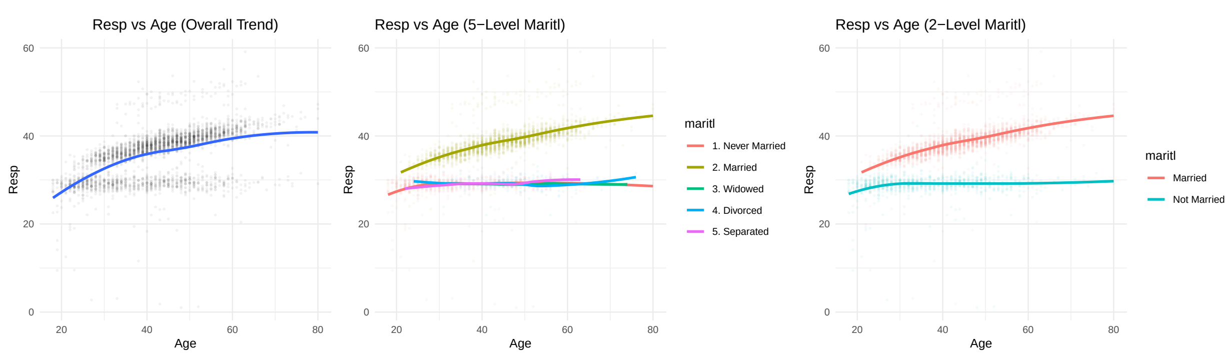
Resp vs log(Wage)

While the model without marital status shows one smooth upward trend, the inclusion of marital status produces two distinct trajectories, especially at higher wage levels, illustrating a meaningful interaction effect.



Resp vs Age

Including marital status reveals an interaction effect, where married individuals display consistently higher and more rapidly increasing response values across age.



4. Modeling Strategy and Results

4.1 Data Partitioning and Modeling Approach

The cleaned data ($n = 2,940$) was partitioned into an **80% training set** (Wage.train) for model fitting and hyperparameter tuning, and a **20% test set** (Wage.test) for unbiased performance evaluation.Our modeling strategy moved from simple parametric models to complex non-parametric methods to find the optimal balance between predictive accuracy (low Test MSE) and statistical inference (interpretability):

- **Baseline & Regularized Models:** (MLR, Ridge, LASSO) provide a simple linear benchmark.
- **Interpretable Non-Linear Models:** (Polynomial Regression, GAMs) capture non-linear trends and key interactions.
- **Non-Parametric Ensemble Models:** (Random Forest, Boosting) offer maximum predictive power by capturing arbitrary interactions.

4.2 Performance of Interpretable Non-Linear Models

4.2.1 Polynomial Regression:

Polynomial regression was first used as a simple and interpretable way to introduce curvature in the model.

- Captures curvature without fully abandoning linear structure
- Used as a first benchmark for nonlinearity
- Fit with and without marital-status interaction terms
- Helps determine whether interaction effects meaningfully improve model accuracy

Polynomial Regression Results

The Best Polynomial Model used a third-degree polynomial for logwage and a second-degree polynomial for age, with both terms fully interacting with maritl (recoded as Married/Not Married), plus all other categorical predictors.

$$\text{Resp} \sim \text{poly}(\text{logwage}, 3) \times \text{maritl} + \text{poly}(\text{age}, 2) \times \text{maritl} + \text{All Categoricals}$$

Table 7: log(Wage) and Age Polynomial Test MSE Results

Model	Test_MSE
Age Polynomial (Degree 2)	19.338723
log(Wage) Polynomial (Degree 3)	18.967037
Age Polynomial (Degree 2) with Marriage Interaction	4.898314
log(Wage) Polynomial (Degree 3) with Marriage Interaction	4.313557
Age and log(wage) Polynomial (no interaction)	13.667923
Age and log(wage) with marital status interaction	1.046312

The results show that adding nonlinear terms for both predictors does not improve performance by much unless interaction effects are included, indicating that subgroup differences by marital status drive much of the predictive gain.

Introducing Categorical Predictors

In an effort to improve the model's performance, the age and log(Wage) model with interaction effects is set as the baseline. From there, various categorical values were tested to gauge an improvement in Test MSE.

Table 8: Top 5 Polynomial Models by Test MSE

Model	Test_MSE
Base + education + jobclass + health_ins	0.973
Base + jobclass + health_ins	0.973
Base + education + jobclass + health + health_ins	0.974
Base + jobclass + health + health_ins	0.975
Base + education + jobclass	0.975

The top-performing polynomial models all include the full or near-full set of categorical predictors, suggesting that demographic and job-related characteristics contribute additional explanatory value beyond nonlinear age and wage effects.

4.2.2 Generalized Addative Models (GAM):

GAMs were implemented to address the limitations of polynomial models by learning smooth, data-driven functions rather than relying on fixed polynomial shapes.

- Allows for smooth functions rather than assuming polynomial shape
- Supports different smooths by marital status, aligning with observed interaction effects
- Provides interpretability while modeling nonlinear patterns more accurately
- Iteratively expanded to include additional categorical predictors (education, race, jobclass)

Generalized Addative Models (GAM) Results:

A Generalized Additive Model (GAM) was then fit to replicate the baseline structure used in the polynomial models, incorporating log(Wage) and Age along with their interactions with marital status. Unlike polynomial regression, GAM allows the use of smoothing spline functions, enabling a flexible, non-parametric fit that can better capture nonlinear patterns without imposing a fixed global shape.

The Best GAM used smoothing splines (s) for logwage and age, with their smooth functions varying based on maritl (known as gam12 in the internal analysis):

$$\text{Resp} \sim s(\text{logwage}, \text{by} = \text{maritl}) + s(\text{age}, \text{by} = \text{maritl}) + \text{All Categoricals}$$

Table 9: Top 5 GAM Models by Test MSE

Model	Test_MSE
Base + education + jobclass + health_ins	0.984
Base + jobclass + health_ins	0.986
Base + jobclass	0.988
Base + jobclass + health + health_ins	0.989
Base + education + jobclass	0.989

Similar to polynomial regression, these results suggest that the GAM framework benefits from a full or nearly full list of additional categorical predictors, which enhance predictive accuracy when paired with spline-based nonlinear effects.

Comparison to Polynomial Regression

Compared to the polynomial regression models, the GAM produced slightly higher Test MSE, likely because its smoother spline-based functions constrain flexibility and introduce bias by not attempting to capture sharp curvature in the data. This slight increase in bias does result in a slightly lower residual variance, as observed in the table below.

Table 10: Comparison of Best Polynomial and GAM Models

Metric	Best.Polynomial	Best.GAM
Test MSE	0.980	0.995
10-fold CV MSE	1.063	1.091
Residual Variance	1.005	1.002
Total Effective DF	40.000	48.680

The best polynomial model achieves slightly lower test MSE (0.976 vs 0.986) and lower 10-fold CV MSE (1.031 vs 1.056) than the GAM, indicating marginally better out-of-sample performance and stability with fewer effective degrees of freedom (22 vs 31.11). Although the GAM attains a very slightly lower residual variance, this comes at the cost of greater complexity, so the polynomial model offers a better balance of accuracy, parsimony, and interpretability.

Tree Based Method Results

Description as to why...

Include Table Here for 4 Methods

- Test MSE
- CV MSE
- Residual Variance
- Effective Degrees of Freedom

Short summary of best one.

Comparison to Parametric Models

Explain pros but mostly cons because it doesn't perform as well

Model Evaluation

- Explain how close the polynomial and GAM are in results (compare bias and variance). Lowkey this is basically the section I did above (Summary Comparison of Best Polynomial and GAM Models) so maybe it would be better here – perhaps a copy paste scenario. You'd have to add the tree results too, just pick the best of the tree methods (boosting I think)
- Looks like Polynomial has the edge in Test MSE and CV MSE. Although it barely loses the variance battle, its effective DF is so much lower than the GAM spline method.
- Talk about explainability of polynomial method vs others.
- Write out final Model Equation

Appendix

Appendix 1

```
##
## Call:
## lm(formula = Resp ~ age * education, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.464  -2.133   1.460   2.786  18.285
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   24.120395   1.006477  23.965  < 2e-16 ***
## age                            0.247916   0.023062  10.750  < 2e-16 ***
## education2. HS Grad            0.967445   1.152390   0.840  0.40125
## education3. Some College      -0.122795   1.220315  -0.101  0.91985
## education4. College Grad       1.348483   1.243922   1.084  0.27843
## education5. Advanced Degree    4.274717   1.445696   2.957  0.00313 **
## age:education2. HS Grad       -0.012724   0.026357  -0.483  0.62930
## age:education3. Some College   0.015126   0.028214   0.536  0.59190
## age:education4. College Grad   0.002141   0.028392   0.075  0.93989
## age:education5. Advanced Degree -0.025325   0.032253  -0.785  0.43241
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.714 on 2930 degrees of freedom
## Multiple R-squared:  0.2984, Adjusted R-squared:  0.2962
## F-statistic: 138.5 on 9 and 2930 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Resp ~ age * jobclass, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.410  -2.284   1.420   2.679  17.964
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)               24.41599    0.45339  53.852   <2e-16 ***
## age                        0.25397    0.01054  24.097   <2e-16 ***
## jobclass2. Information     1.50252    0.67798   2.216   0.0268 *
## age:jobclass2. Information -0.01231    0.01539  -0.800   0.4238
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.786 on 2936 degrees of freedom
## Multiple R-squared:  0.2756, Adjusted R-squared:  0.2748
## F-statistic: 372.3 on 3 and 2936 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Resp ~ age * health, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -35.719  -2.321   1.305   2.755  18.574
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)             25.06366    0.62412  40.159  < 2e-16 ***
## age                      0.22823    0.01338  17.051  < 2e-16 ***
## health2. >=Very Good    -0.66872    0.74268  -0.900  0.36798
## age:health2. >=Very Good  0.04718    0.01635   2.885  0.00394 **
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.763 on 2936 degrees of freedom
## Multiple R-squared:  0.2824, Adjusted R-squared:  0.2816
## F-statistic:   385 on 3 and 2936 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Resp ~ age * maritl, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.432  -0.974  -0.085   0.824  16.549
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             27.451294   0.368393  74.516  < 2e-16 ***
## age                      0.042513   0.010764   3.949 8.02e-05 ***
## maritl2. Married         1.579807   0.461416   3.424 0.000626 ***
## maritl3. Widowed         1.671434   3.059256   0.546 0.584865
## maritl4. Divorced        1.579128   1.060950   1.488 0.136751
## maritl5. Separated      -0.121284   1.766523  -0.069 0.945267
## age:maritl2. Married     0.172202   0.012341  13.954  < 2e-16 ***
## age:maritl3. Widowed    -0.043987   0.057946  -0.759 0.447853
## age:maritl4. Divorced   -0.042042   0.023153  -1.816 0.069493 .
## age:maritl5. Separated   0.001752   0.039946   0.044 0.965014
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.81 on 2930 degrees of freedom
## Multiple R-squared:  0.7508, Adjusted R-squared:  0.7501
## F-statistic: 980.9 on 9 and 2930 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Resp ~ age * health_ins, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.222  -2.315   1.338   2.704  18.296
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          26.246081   0.433896  60.489  < 2e-16 ***
## age                   0.231193   0.009672  23.903  < 2e-16 ***
## health_ins2. No      -2.799121   0.687987  -4.069 4.85e-05 ***
## age:health_ins2. No   0.037786   0.016010   2.360   0.0183 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.773 on 2936 degrees of freedom
## Multiple R-squared:  0.2794, Adjusted R-squared:  0.2786
## F-statistic: 379.4 on 3 and 2936 DF,  p-value: < 2.2e-16
```

Appendix 2

```
##
## Call:
## lm(formula = Resp ~ logwage * education, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.4318  -3.4202   0.4979   3.3329  14.5442
##
## Coefficients:
```

```
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        -7.5057     4.5991  -1.632   0.1028
## logwage                             9.5466     1.0439   9.146   <2e-16 ***
## education2. HS Grad                 0.1344     5.1380   0.026   0.9791
## education3. Some College          -11.6923     5.3420  -2.189   0.0287 *
## education4. College Grad            5.1910     5.2524   0.988   0.3231
## education5. Advanced Degree        -6.7308     5.6107  -1.200   0.2304
## logwage:education2. HS Grad        -0.1630     1.1599  -0.141   0.8883
## logwage:education3. Some College    2.0911     1.1966   1.748   0.0807 .
## logwage:education4. College Grad   -1.4812     1.1709  -1.265   0.2060
## logwage:education5. Advanced Degree  1.0683     1.2280   0.870   0.3844
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.556 on 2930 degrees of freedom
## Multiple R-squared:  0.3448, Adjusted R-squared:  0.3428
## F-statistic: 171.4 on 9 and 2930 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Resp ~ logwage * jobclass, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.5334  -3.4330   0.4929   3.4276  13.9825
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   -5.4762     1.6410  -3.337 0.000857 ***
## logwage                        8.8189     0.3572  24.689  < 2e-16 ***
## jobclass2. Information        -1.6739     2.3052  -0.726 0.467812
## logwage:jobclass2. Information   0.3954     0.4941   0.800 0.423594
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.618 on 2936 degrees of freedom
## Multiple R-squared:  0.3253, Adjusted R-squared:  0.3246
## F-statistic: 471.8 on 3 and 2936 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Resp ~ logwage * health, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.2496  -3.3901   0.4582   3.3853  13.6121
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 -7.6387     2.1353  -3.577 0.000353 ***
## logwage                      9.4104     0.4663  20.180  < 2e-16 ***
## health2. >=Very Good         0.5992     2.5239   0.237 0.812358
## logwage:health2. >=Very Good  -0.2760     0.5471  -0.505 0.613928
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.61 on 2936 degrees of freedom
## Multiple R-squared:  0.3278, Adjusted R-squared:  0.3272
## F-statistic: 477.4 on 3 and 2936 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Resp ~ logwage * maritl, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -26.1661  -1.5550  -0.0425   1.5210  13.9609
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 4.0400     1.4291   2.827  0.00473 **
## logwage                     5.5400     0.3184  17.400  < 2e-16 ***
## maritl2. Married            5.3401     1.6461   3.244  0.00119 **
## maritl3. Widowed           18.3466    13.3674   1.372  0.17002
## maritl4. Divorced          -4.8871     2.9775  -1.641  0.10084
## maritl5. Separated          8.1478     6.2599   1.302  0.19316
## logwage:maritl2. Married    0.6660     0.3622   1.839  0.06605 .
## logwage:maritl3. Widowed   -4.0931     2.9020  -1.410  0.15852
## logwage:maritl4. Divorced   0.9841     0.6516   1.510  0.13104
## logwage:maritl5. Separated -1.8076     1.3670  -1.322  0.18616
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.67 on 2930 degrees of freedom
## Multiple R-squared:  0.775,  Adjusted R-squared:  0.7743
## F-statistic:  1121 on 9 and 2930 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = Resp ~ logwage * health_ins, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4380  -3.4463   0.4241   3.3726  14.1796
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)             -3.6252     1.5696  -2.310    0.021 *
## logwage                  8.4247     0.3305  25.494  < 2e-16 ***
## health_ins2. No        -10.0078     2.4300  -4.118 3.92e-05 ***
## logwage:health_ins2. No   2.3026     0.5302   4.343 1.46e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.6 on 2936 degrees of freedom
## Multiple R-squared:  0.3308, Adjusted R-squared:  0.3301
## F-statistic: 483.7 on 3 and 2936 DF,  p-value: < 2.2e-16
```

Appendix 3

```
## Analysis of Variance Table
##
## Model 1: Resp ~ age + maritl
## Model 2: Resp ~ age * maritl
##   Res.Df   RSS Df Sum of Sq      F    Pr(>F)
## 1   2934 25331
## 2   2930 23129  4      2202 69.738 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
## Model 1: Resp ~ age + health
## Model 2: Resp ~ age * health
##   Res.Df   RSS Df Sum of Sq      F   Pr(>F)
## 1   2937 66801
## 2   2936 66612  1    188.85 8.3237 0.003942 **
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
## Model 1: Resp ~ age + health_ins
## Model 2: Resp ~ age * health_ins
##   Res.Df   RSS Df Sum of Sq      F  Pr(>F)
## 1   2937 67014
## 2   2936 66887  1     126.9 5.5703 0.01833 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Appendix 4

```
## Analysis of Variance Table
##
## Model 1: Resp ~ logwage + health_ins
## Model 2: Resp ~ logwage * health_ins
##   Res.Df   RSS Df Sum of Sq      F    Pr(>F)
## 1   2937 62518
## 2   2936 62119  1    398.99 18.858 1.456e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
## Model 1: Resp ~ logwage + maritl
## Model 2: Resp ~ logwage * maritl
##   Res.Df   RSS Df Sum of Sq      F  Pr(>F)
## 1   2934 20957
## 2   2930 20888  4    68.884 2.4157 0.04675 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Appendix 5

Appendix 6

Appendix 7

Appendix 8

Appendix 9

Appendix 10

Appendix 11